Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/19/2019
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	Membrane Contacts in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/29/2018	**End Date:**	10/28/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0145		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants grown during spaceflight experience conditions of microgravity, ionizing radiation, and reduced gas exchange. If we are to use plants in advanced life-support systems during long space missions, it is imperative that we understand how plants respond to these varying conditions to ensure their productivity. Plant growth on Earth is shaped in great part by the force of gravity, while conditions of microgravity experienced in space result in dramatic changes in organ orientation and growth. The plant response to gravity, or gravitropism, is also important for the growth angle of root and shoot branches, and therefore, it also impinges in crop architecture and productivity on Earth. The effects of microgravity on plant growth at the cellular level is not yet fully characterized, and the molecular mechanisms by which plants perceive gravity are still unknown. A wealth of state-of-the-art genetic, imaging, and omics tools in the model plant Arabidopsis thaliana are ripe to provide the clues for this elusive system in plant biology, and novel players have recently been identified. We propose here to characterize the role of two protein families predicted to mediate contacts between the endoplasmic reticulum (ER) and other organelles as candidates for important roles in gravitropism. These proteins were selected based on our previous NASA-funded research where we characterized the role of vacuoles in gravitropism. This work led us to propose the hypothesis that membrane contacts between amyloplasts and vacuoles and/or ER have a role in gravity sensing in shoot statocytes. We plan to test this hypothesis directly by addressing the specific aims to 1) determine the function of candidate membrane tethers in membrane contacts and gravity sensing; and 2) determine the effects of spaceflight on vacuole fusion and ER morphology. A reverse genetic approach combined to fluorescence microscopy and mutant characterization will be used to characterize the role of selected tethers. The flight experiment will use well-established methods for Arabidopsis growth in Biological Research in Canister-Petri Dish Fixation Unit (BRIC-PDFU) units that are suitable for chemical treatment and fixation. Plant sub-cellular phenotypes will be analyzed after sample return to Earth. Our ground-based research will address the most important knowledge gap in gravitropism at the cellular level and in specific cell types of the Arabidopsis shoot. In addition, the proposed flight experiment will result in multiple outcomes beyond documenting the effects of spaceflight on organelle fusion. These include probing the use of chemical inhibitors for plant cell biology experimentation in space and testing recently described tissue clearing protocols for visualization of cellular organelles in space-grown plants. Our research will train young scientists in space biology research including undergraduate students, and therefore contribute to the development of human resources in STEM (Science, Technology, Engineering, and Math).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. The identification of candidate proteins involved in membrane contacts and gravitropism will enhance the fundamental understanding of how plants perceive and respond to gravity signals.
Task Progress:	We have made progress on genotyping of T-DNA insertion lines from candidate proteins and testing for gravitropism defects. Seven different candidate genes have been tested so far but only one has defects on gravitropism. As we gear towards our flight experiment, we have been testing all the conditions of storage length and temperature for the chemicals that will be used in the International Space Station (ISS). We have found no decrease in activity for these molecules after dilution and storage.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Abstracts for Journals and Proceedings	McLaurin C, Rojas-Pierce M. "Assessing the role of membrane contact proteins in plant shoot gravitropism." Presented at the North Carolina State University Summer Undergraduate Research Symposium, Raleigh, NC, August 2019. North Carolina State University Summer Undergraduate Research Symposium, Raleigh, NC, August 2019. , Aug-2019
Abstracts for Journals and Proceedings	Ly V, Selwanes J, Brillada C, Danz K, Tomlinson T, Rojas-Pierce M. "Are membrane contacts between the vacuole, ER, and amyloplasts important for shoot gravitropism?" Poster presented at the Midwest Plant Cell Dynamics Meeting, State College, PA, June 18-21, 2019. Midwest Plant Cell Dynamics Meeting, State College, PA, June 18-21, 2019. , Jun-2019